

MPI

Ch-6: Stack & Subroutines, Counters & Time Delay, Code Conversion, BCD Arithmetic & 16-bit Operations

Stack & Subroutines

Year	Question No.	Marks	Question
Summer 2024	Q.4(a)	03	Define Stack. Explain PUSH and POP instructions.
Winter 2024	Q.4(a)	03	Define the concepts of stack and subroutines in the context of the 8085 microprocessor.
Summer 2025	Q.4(a)	03	Define (1) Stack (2) Counter (3) Time Delay.
Summer 2025	Q.4(b)	04	Explain subroutine in 8085 Microprocessor with example.
Winter 2022	Q.4(a)	03	State the difference between PUSH and POP instruction.
Winter 2021	Q.3(a)	03	Explain subroutine with suitable example.

Counters & Time Delay Routines

Year	Question No.	Marks	Question
Summer 2024	Q.3(c)	07	Explain counters and time delay with suitable example. Also specify various applications of counters and time delay.
Summer 2023	Q.2(c)	07	If an 8085 microprocessor has a 2 MHz crystal frequency, what is the

Year	Question No.	Marks	Question
			assembly language code required creating a delay of 1 second?
Winter 2024	Q.4(c)	07	Create an 8085-assembly language routine to implement a delay, utilizing both counters and the stack.
Winter 2022	Q.4(c)	07	Write a program to count continuously in hexadecimal from FFH to 00H in a system with a clock period of 0.5 μ s. Use register C to set up 1 millisecond delay between each count and display the number at the output port1.
Winter 2022	Q.2(c) OR	07	Write an assembly language program to provide the given ON/OFF time to traffic lights and pedestrian signs. Also write Delay subroutine to generate appropriate delay.
Winter 2021	Q.4(b)	04	What amount of delay is generated if the crystal frequency is 4 MHz? (Program based delay calculation)
Summer 2025	Q.3(c) OR	07	Write a set of 8085 assembly language instructions to generate a 1 second delay, if the crystal frequency is 1 MHz.

Code Conversion / BCD Arithmetic

Year	Question No.	Marks	Question
Winter 2021	Q.4(b) OR	04	Write an 8085 assembly language program to convert a two-digit BCD number into its equivalent hexadecimal number.
Summer 2022	Q.2(a)	03	Explain the following instructions: (1) LHLD (2) RAL (3) DAA.

16-Bit Data Operations

Year	Question No.	Marks	Question
Summer 2025	Q.3(a)	03	Explain function of given instructions: (1) DAD (2) LHLD (3) XCHG.
Winter 2021	Q.3(b)	04	Explain following instructions with no. of bytes, machine cycles and T-states required for execution: (1) SHLD (2) RAL.
Winter 2022	Q.3(c) OR	07	Specify the addressing mode, required Machine cycles, T-States and function for following instructions: 1. MVI M, 45H 2. RAL 3. LHLD 2300H

Ch-7: Interfacing Concepts, Ports, Interrupts, 8259A, 8255A

Interfacing Concepts / Ports / I/O Devices

Year	Question No.	Marks	Question
Summer 2024	Q.4(a) OR	03	Differentiate IO-mapped IO and Memory-mapped IO.
Summer 2023	Q.4(b)	04	Compare the memory-mapped IO with the standard IO-mapped IO.
Summer 2022	Q.3(b)	04	Compare memory mapped I/O and I/O mapped I/O.
Winter 2023	Q.4(b) OR	04	Give comparison of Memory mapped I/O and Peripheral mapped I/O.
Winter 2024	Q.4(a)	03	Define I/O ports and describe their role in the process of microprocessor interfacing.

Interrupts in 8085

Year	Question No.	Marks	Question
Summer 2024	Q.4(b)	04	Explain classification of Interrupts in 8085 Microprocessor.
Summer 2023	Q.4(a) OR	03	List the Software and Hardware interrupts of 8085?
Summer 2022	Q.3(c)	07	What are interrupts? List and explain the interrupt available in microprocessor 8085?
Winter 2023	Q.5(a)	03	Differentiate between maskable and non-maskable interrupts.
Winter 2023	Q.5(b) OR	04	How many interrupts are there in 8085? Name them. Explain the characteristics in terms of maskability, vectoring and priority.
Winter 2022	Q.4(b)	04	Differentiate between maskable and non-maskable interrupts.
Winter 2022	Q.4(b) OR	04	What is vectored and non-vectored interrupts?
Winter 2021	Q.3(a) OR	03	What is interrupt? List hardware interrupts of 8085.
Winter 2024	Q.4(c)	07	Discuss the interrupt architecture of the 8085 microprocessor, focusing on both hardware and software interrupt types.
Summer 2025	Q.4(a) OR	03	Differentiate maskable and non-maskable interrupts in 8085 Microprocessor.

Programmable Interrupt Controller 8259A

Year	Question No.	Marks	Question
Summer 2024	Q.4(c) OR	07	Draw and explain block diagram of 8259A.

Year	Question No.	Marks	Question
Summer 2023	Q.4(c)	07	Can you provide an example program in assembly language that initializes a single 8259 Programmable Interrupt Controller (PIC) connected to an 8085 processor?
Summer 2022	Q.4(c)	07	Draw the internal block diagram of 8259A and explain the functions of each block in detail.
Winter 2024	Q.4(b) OR	04	Describe how the 8259A Programmable Interrupt Controller functions and how it is utilized in microprocessor systems.
Winter 2022	Q.5(c)	07	Explain the internal Block diagram of 8259A.
Winter 2021	Q.5(c) OR	07	Draw and explain the block diagram of the programmable interrupt controller 8259A.
Summer 2025	Q.4(c) OR	07	Explain Programmable Interrupt Controller 8259A.

Programmable Peripheral Interface 8255A

Year	Question No.	Marks	Question
Summer 2024	Q.4(c)	07	Draw and explain block diagram of 8255A.
Summer 2024	Q.4(b) OR	04	Explain BSR Mode in 8255A.
Summer 2023	Q.4(b) OR	04	Discuss the operation of the 8255 Programmable Peripheral Interface and how it is interfaced with an 8085 microprocessor. What are the different modes of operation and how are they selected?
Summer 2022	Q.5(c)	07	Draw the internal block diagram of 8255 and explain the functions of

Year	Question No.	Marks	Question
			each block in details.
Winter 2024	Q.4(b)	04	Explain the working of the 8255 Programmable Peripheral Interface.
Winter 2022	Q.5(c) OR	07	Explain the internal Block diagram of 8255A.
Winter 2021	Q.5(c)	07	Draw and explain the block diagram of the programmable peripheral interface 8255A.
Summer 2025	Q.4(c)	07	Draw and explain input and output handshake mode of 8255A.
Summer 2025	Q.4(b) OR	04	Explain control register in 8255A with diagram.

Ch-8: Advanced Microprocessors (8086, 80286, 80386)

8086 Logical Block Diagram / Architecture / Pin Functions

Year	Question No.	Marks	Question
Summer 2024	Q.5(c)	07	Draw and explain logical block diagram of 8086 Microprocessor.
Summer 2023	Q.5(c)	07	Draw and explain internal architecture and PIN diagram of 8086 microprocessor.
Summer 2022	Q.5(a) OR	03	Explain the given pins of 8086: (1) ALE (2) DEN (3) MN/MX.
Summer 2022	Q.5(c) OR	07	Explain the block diagram of 8086 microprocessor.
Winter 2024	Q.5(a) OR	03	Describe the pin configuration of the 8086 microprocessor in detail.

Year	Question No.	Marks	Question
Summer 2025	Q.5(c) OR	07	Draw and explain pin diagram of 8086 Microprocessor.

Segmentation / Segment Registers / Segment Types

Year	Question No.	Marks	Question
Summer 2024	Q.5(b) OR	04	Explain concepts of segmentation in 8086 Microprocessor.
Summer 2023	Q.5(b) OR	04	How the physical addresses are calculated from segment register in 8086 microprocessor?
Winter 2024	Q.5(a)	03	Describe the concept of segmentation in the 8086 microprocessor.
Winter 2021	Q.5(a) OR	03	List and explain the segment registers of 8086 microprocessor.
Summer 2025	Q.5(b)	04	Explain segment registers in 8086 Microprocessor.

Minimum & Maximum Mode

Year	Question No.	Marks	Question
Summer 2022	Q.5(b) OR	04	Explain the modes of operation of 8086 microprocessor.
Winter 2024	Q.5(b)	04	Compare the minimum mode and maximum mode of the 8086 microprocessor.
Summer 2025	Q.5(a)	03	Describe operating modes in 8086 Microprocessor.

80286 Architecture / Programming Model / Memory Management

Year	Question No.	Marks	Question
Summer 2024	Q.5(b)	04	Explain register organization of 80286 Microprocessor.
Summer 2023	Q.5(a)	03	Differentiate 80286 with 80386 microprocessor.
Summer 2023	Q.5(a) OR	03	Differentiate between the real mode and protected mode of the 80286 microprocessor.
Summer 2023	Q.5(c) OR	07	Describe the architecture of the 80286 with a neat block diagram.
Summer 2024	Q.5(a) OR	03	Describe protected virtual address mode in 80286 Microprocessor.
Winter 2024	Q.5(b) OR	04	How does the 80286 microprocessor improve upon the 8086, particularly in terms of memory management?
Winter 2024	Q.5(c) OR	07	Summarize the programming model and data types available in the 80286 microprocessor.
Winter 2021	Q.5(b)	04	Draw block diagram of 80286 microprocessor.
Summer 2025	Q.5(c)	07	Draw and explain architecture of 80286 Microprocessor.
Summer 2025	Q.5(a) OR	03	Differentiate GDT and LDT in 80286 Microprocessor.

80386 Architecture / Programming Model / Descriptor

Year	Question No.	Marks	Question
Summer 2024	Q.5(c) OR	07	Draw and explain architecture of 80386 Microprocessor.
Winter 2024	Q.5(c)	07	Summarize the architecture and programming model of the 80386 microprocessor.

Year	Question No.	Marks	Question
Winter 2021	Q.5(a)	03	Explain format of the descriptor in 80386 with diagram.
Winter 2021	Q.5(b) OR	04	Draw block diagram of 80386 microprocessor.
Summer 2025	Q.5(b) OR	04	Explain register organization in 80386 Microprocessor.